DTC P0087:00 [SKYACTIV-D 1.5]

id0102q2008200

DTC P0087:00	Low fuel pressure malfunction in common rail fuel pressure control system
DETECTION CONDITION	• With the following conditions met, a condition has continued for **15 s** in which the fuel pressure sensor output value is lower than the fuel pressure control target value by **20 MPa {204 kgf/cm^2, 2901 psi}**: — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Fuel pressure control: Being performed • Change in target fuel pressure during half-rotation of crankshaft: **−6 to 6 MPa {−61—61 kgf/cm^2, −870—870 psi}** • Fuel temperature: **−25—70 °C {−13—158 °F}** • Target fuel discharge amount from supply pump: **Less than 10,000 mm^3/rev** — The following DTCs are not detected: • Fuel temperature sensor: P0181:00, P0182:00, P0183:00 • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 • Suction control valve: P062A:00 **Diagnostic support note** • This is an intermittent monitor (fuel system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Increase the idle speed. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Fuel filter clogged • Fuel pressure relief valve malfunction • Suction control valve malfunction • Fuel pressure sensor malfunction • Fuel pipe perforation or breakage (improper connection) • Fuel leakage from fuel line or clogging • Supply pump malfunction • Poor fuel quality • Fuel is frozen • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (fuel system related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT FUEL FILTER** • Inspect the fuel filter for clogging. (See FUEL FILTER INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 12. (See FUEL FILTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See FUEL FILTER DISASSEMBLY/ASSEMBLY [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**INSPECT FUEL PRESSURE RELIEF VALVE** • Inspect the fuel pressure relief valve. (See FUEL PRESSURE RELIEF VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the common rail, then go to Step 12. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**INSPECT SUCTION CONTROL VALVE** • Inspect the suction control valve. (See SUCTION CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the suction control valve, then go to Step 12. (See SUCTION CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**INSPECT FUEL PRESSURE SENSOR** • Inspect the fuel pressure sensor. (See FUEL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the common rail, then go to Step 12. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**INSPECT FUEL PIPE** • Inspect the fuel pipe installation condition. (See FUEL SYSTEM LOCATION INDEX [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Install the fuel pipe properly, then go to Step 12.
		No	Go to the next step.
9	**INSPECT FOR FUEL LEAKAGE FROM FUEL LINE OR CLOGGING** • Inspect the following fuel line for fuel leakage or clogging. — Between supply pump and common rail — Between common rail and fuel injector • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 12.
		No	Go to the next step.
10	**INSPECT SUPPLY PUMP** • Inspect the supply pump. (See SUPPLY PUMP INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the supply pump, then go to Step 12. (See SUPPLY PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
11	**INSPECT FOR MALFUNCTION DUE TO POOR FUEL** • Replace the fuel. (See FUEL DRAINING PROCEDURE [SKYACTIV-D 1.5].) • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Access the ECT PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Start the engine and idle it. • Wait until the ECT PID value is **above 80 °C {176 ° F}**. • Wait for **1 min** (idle). • Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE for this DTC present?	Yes	Go to the next step.
		No	Advise the customer as to the change in the fuel used. Go to Step 13.
12	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Access the ECT PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Start the engine and idle it. • Wait until the ECT PID value is **above 80 °C {176 ° F}**. • Wait for **1 min** (idle). • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
13	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.